

the brushless DC motors segment lately. These motors are very useful for building robots, as these do not wear out easily. Some of the recently-released components are FAULHABER 3274 BP4 series DC servomotors, FAULHABER 2057XXXXBHS series brushless DC servo motors, Nanotec Electronic's DB80, Maxon Motor's EC-4pole 30 and Global Electric Motor Solutions' NEMA hybrid stepper motors, to name a few.

Onboard gyroscopes, accelerometers, magnetometers, temperature and current sensors are some other components required to monitor the health and location of the robot, informs Sinha. At times, all these components come integrated in a board. "For instance, Pixhawk auto pilot already has inbuilt gyroscope, accelerometer and magnetometer, while the global positioning system (GPS), temperature and current sensors can be additionally incorporated," says Sinha. "Off-the-shelf components can be bought with a lot of ease today than a few years ago," he adds.

There are certain key aspects that one has to consider while developing

SAM—the brick-laying robot

Designed by Construction Robotics, a company based in New York, USA, semi-automated mason (SAM) is a robotic brick-laying machine, which looks to increase productivity of human masons by assisting them in laying bricks. While the robotic arm performs mechanical tasks like lifting bricks, applying cement and placing these as programmed, the mason takes care of the aesthetic details. The company claims that SAM can lay 800 to 1200 bricks, whereas a mason, in comparison, can lay 300 to 500 bricks a day.



SAM in action

highly-interactive robots. Cognitive intelligence, omni-directional mobil-

ity, gravity-compensated manipulation and back drivability are some of these, informs Jayakrishnan T., CEO, ASIMOV Robotics Pvt Ltd. He says, "The new concept of flexible manufacturing is a typical example. Hence, most of our robot platforms are either compatible or come with these capabilities integrated into them."

Availability of components is a challenge as many components like sensors and motors from reliable vendors are not easily available in India. "Steadily, we are witnessing a network of distributors emerging in India who import these components and make these available for robot makers. Some examples of such distributors are Robokits and RcBazaar," informs Sinha.

High degree of standardisation required

Standards is one aspect lacking in the robotics industry, feel experts. There have been some success-

Point of view: Testing and manufacturing of robotic designs

"As far as testing is concerned, we take a multi-modal approach. For most part, we are able to debug in software. While doing this, we can predict the functioning of particular pieces of hardware. If there are major issues or errors, we make use of oscilloscopes to perform signal analysis, but this approach is diminishing as the hardware is getting more reliable nowadays. We use some of the advanced oscilloscopes by Anritsu to debug major faults. Although expensive, these ease the process of tracking a major bug in a design.



Aakash Sinha, CEO and founder, Omnipresent Robot Technologies Pvt Ltd

"Manufacturing boards in India is generally hassle-free. There is a good ecosystem of designing, creating and mounting printed circuit boards (PCBs). On the downside, there are no reliable one-stop vendors that we have come across. There are very few players willing to create a one-stop ecosystem where we share the components list and the design, while they create a finished tested board. An ideal one-stop shop would work on printing PCBs, arranging for components, mounting and testing these. For our needs, we just identify players who consistently use electronics manufacturing equipment and outsource the post-design work."